

Chemical and Biomolecular Engineering Program
CHE 450 Syllabus – Fall 2024

1. Department, Number, and Course Title: CHE 450 Chemical Engineering Design I
2. Class Type, Time, Days & Location: *Lecture* 10:15 – 11:30 AM, T Th, 1025 EB2
3. Instructor:

Dr. Matthew Cooper (he / him / his)

Office: 2016 EB1, Phone: 513-1623, Email: mecoope3@ncsu.edu

Office Hours (held via Zoom, link on Moodle site): W F 10:00 – 11:30 AM

4. Teaching Assistants and Graders:

Teaching Assistants

Morgan Hurst

Email: mrrhurst@ncsu.edu

Office Hours (in EB1 Atrium, Zoom upon request): M 1:00 – 4:00 PM

Belinda Mativenga

Email: bymative@ncsu.edu

Office Hours (in Plant Sciences 3310, Zoom upon request): T Th 1:30 – 3:00 PM

Grader

Triveni Sai Manisha Koyya

Email: tkoyya@ncsu.edu

5. Course (Catalog) Description: Applications of cost accounting, cost estimation for new equipment, manufacturing cost and measures of profitability. Use of computer simulation design and cost programs. Procedures for sizing unit operations commonly encountered in the chemical process industry. Heuristics for selection of separation processes and heat exchanger network synthesis.
6. Prerequisite(s): Prerequisite: CHE 312; Corequisite: CHE 446 or CHE 447
7. Textbook(s) and/or Other Required Material: Turton, Shaeiwitz, Bhattacharyya and Whiting, Analysis, Synthesis and Design of Chemical Processes, 5th Edition (Prentice Hall), ISBN 978-0134177403, Cost: \$180
8. Course Learning Objectives:
 - 1) Apply and synthesize information learned in previous courses toward the design and/or operation of chemical processes
 - 2) Apply concepts and calculations related to time value of money, cost components for plant economics such as capital cost and manufacturing cost estimations, and/or typical economic analysis methods such as net present value and return on investment
 - 3) Use ASPEN to model typical chemical engineering unit operations

- 4) Develop team oriented performance criteria, problem-solving skills, and assessment processes
- 5) Incorporate process safety considerations into process design
9. Class/Laboratory Schedule (# sessions/week and session duration): Two 75-minute lectures per week.
10. Curricular Course Contribution: Engineering Topics (3 semester hours) with design content
11. Scheduled Assignment/Exam Dates

Date	Location and Time	Assignment Due
August 27 (Tues)	Moodle, 10:15 AM	CHE 450/451 Senior Design Project Survey Feedback
September 5 (Thurs)	Moodle, 10:15 AM	Homework #1
September 12 (Thurs)	Moodle, 10:15 AM	Homework #2
September 19 (Thurs)	TurnItIn.com, 10:15 AM	<u>Project Deliverable</u> : Team Contract
September 24 (Tues)	Moodle, 10:15 AM	Homework #3
September 24 (Tues)	Outside of Class, 11:30 AM	Contents Under Pressure Gameplay Window Begins
October 1 (Tues)	1025 EB2, 10:15 AM	Exam #1
October 8 (Tues)	TurnItIn.com, 10:15 AM	<u>Project Deliverable</u> : Technical Literature Review
October 10 (Thurs)	Moodle, 10:15 AM	Homework #4
October 17 (Thurs)	Moodle, 10:15 AM	Homework #5
October 20 (Sun)	Outside of Class, 11:59 PM	Contents Under Pressure Gameplay Window Ends
October 24 (Thurs)	Moodle, 10:15 AM	Homework #6
October 31 (Thurs)	Moodle, 10:15 AM	Homework #7
November 5 (Tues)	TurnItIn.com, 10:15 AM	<u>Project Deliverable</u> : Project Plan and Schedule
November 7 (Thurs)	Moodle, 10:15 AM	Homework #8
November 14 (Thurs)	Moodle, 10:15 AM	Homework #9
November 21 (Thurs)	1025 EB2, 10:15 AM	Exam #2
November 26 (Tues)	Moodle, 10:15 AM	Homework #10
December 3 (Tues)	TurnItIn.com, 11:59 PM	<u>Project Deliverable</u> : Fall Project Status Report

12. Grading

41% Project Deliverables: Team Contract (5%), Technical Literature Review (12%), Project Plan and Schedule (12%), Fall Project Status Report (12%)

5% Process Safety Video Game Completion

20% 10 Homework Sets (2% each)

34% 2 Exams (17% each)

An individual's score (IS) on a project deliverable (specifically the team contract, literature review, project plan and schedule, and fall project status report) is determined based on CATME peer evaluations and the average CATME team score (TS) using the following formula:

$$IS_n = TS(0.50 + 0.50 \frac{(\sum_{i=1}^n \frac{IPE_{n,i} - IPE_{n,n}}{APE_i^* - APE_n})}{n-1}) \quad (1)$$

where, IS_n = individual score of the n^{th} member, $IPE_{n,i}$ = average peer evaluation score for the n^{th} member by the i^{th} member, $IPE_{n,n}$ = average peer evaluation score for the n^{th} member by themselves, APE_i^* = average peer evaluation score by the i^{th} member excluding the member's evaluation of themselves, and APE_n^* = average peer evaluation score by the n^{th} member by themselves

For example: The first and second group member's score (in a five-person team) would be calculated as follows:

$$IS_1 = TS(0.50 + 0.50 \left(\frac{IPE_{1,2} + IPE_{1,3} + IPE_{1,4} + IPE_{1,5}}{APE_2^* + APE_3^* + APE_4^* + APE_5^*} \right) \left(\frac{1}{5-1} \right)) \quad (2)$$

$$IS_2 = TS(0.50 + 0.50 \left(\frac{IPE_{2,1} + IPE_{2,3} + IPE_{2,4} + IPE_{2,5}}{APE_1^* + APE_3^* + APE_4^* + APE_5^*} \right) \left(\frac{1}{5-1} \right)) \quad (3)$$

IPE scores for each team member are awarded by the instructor(s) based on review of the individual's team peer evaluation information.

Course grades will be determined as follows:

Score	>96.5	92.5 to 96.4	89.5 to 92.4	86.5 to 89.4	82.5 to 86.4	79.5 to 82.4	76.5 to 79.4	72.5 to 76.4	69.5 to 72.4	66.5 to 69.4	62.5 to 66.4	59.5 to 62.4	<59.5
Grade	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-	F

Note: grades are not "curved" in this course. It is theoretically possible for everyone in the class to get an A (or an F). Your performance depends only on how you do, not on how everyone else in the class does. It is therefore in your best interests to help your classmates, while acting within the bounds of the stated academic integrity policy.

13. Additional Course Information

- Homework format is specified as follows (format is the similar to that used in CHE 205). Points will be deducted from any homework submissions which do not follow this format:
 - Write on one side of each page
 - Write your name and the problem set number at the top of the first page of your submission.
 - Begin each problem on a new page

- d. Show all equations / calculations
 - e. Box all final answers
 - f. The problems should be submitted in the same order as in the homework assignment
- Homework and some exam materials may be submitted in this class electronically by submitting a scanned copy of your work via the Gradescope link provided on the course Moodle site. Note the following policies; lack of adherence to any of these policies will result in point deductions:
 - a. Submissions must be high-quality scans (not cell phone pictures) in order to ensure legibility for grading. Be aware that typical “scanning” apps that come with iPhones, etc. are often not suitable. It is suggested that students download and use the free CamScanner app for Gradescope submissions; other options are used by students at their own risk of point deductions.
 - b. When submitting scanned materials through Gradescope, submit your problems and any subparts in the correct order. Be sure to select the correct pages in your submission for each question. A tutorial showing how to adhere to these policies is provided in the Gradescope Student Video Tutorial posted to the Moodle site.
 - c. For ASPEN problems, screen shots should be generated using an embedded screen capture method (such as the "Snipping Tool" for PCs or Shift + Command + 4 for Macs). Screenshots which aren't clear, taken with a cell phone, etc. will result in point deductions.
 - d. For ASPEN problems, provide the actual answer to the question being asked as part of your submission (in other words, don't expect the grader to search for it on your screenshot); points will be deducted if the answer to a question is not made explicit.
- Project deliverables will be submitted electronically in both of these two ways: (1) delivered to your project mentor(s) via email or other means requested by your mentor, and (2) by uploading one PDF (preferred format) or MS Word file of the deliverable to the course's TurnItIn.com website by the due time / date specified in the course syllabus for the submission. Instances of plagiarism will be penalized on a situational basis and may be referred to the Office of Student Conduct.
- Any delay in report submission will decrease the report grade by 2% (and an additional 2% per each additional hour of delay, up to a maximum of 10% per calendar day). Reports, etc. may be turned in electronically on weekends/holidays to avoid further penalty.
- Some capstone project mentors prefer that project deliverables are submitted in their “final” state by the due date. Other capstone project mentors prefer that drafts of project deliverables are submitted in advance so they can provide feedback for the team to improve the final draft of the document submitted by the due date. In an effort to be equitable in grading in light of these different mentor preferences, if a mentor asks for a team to submit a draft of a deliverable in advance of the due date, on which they will provide feedback for the team to incorporate into a final draft, both drafts of the report will be graded, and the final team score on the deliverable will be an average of the

scores on the initial and revised reports. This policy applies to draft deliverables submitted in whole or in part to the mentor for feedback before final submission. It should be made clear that whether to grade and provide feedback on grade draft deliverables provided in advance of the due date is entirely the choice of the mentor, not the team.

- Homework is due at the beginning of class; late submissions are penalized 10%. Each additional 24 hours of delay after the corresponding deadline will decrease the grade by an additional 10%. Homework submitted more than 48 hours late will not be accepted.
- A free online response software called Top Hat (www.tophat.com) is a required resource for this class. You will be able to submit answers to in-class questions using Apple or Android smartphones and tablets, laptops, or through text message. Any “clicker” answers will not necessarily be counted as right/wrong, but instead will be used to engage the class, take attendance and/or count for participation. You can visit the Top Hat Overview (<https://success.tophat.com/s/article/Student-Top-Hat-Overview-and-Getting-Started-Guide>) within the Top Hat Success Center which outlines how you will register for a Top Hat account, as well as providing a brief overview to get you up and running on the system. Should you require assistance with Top Hat at any time, due to the fact that they require specific user information to troubleshoot these issues, please contact their Support Team directly by way of email (support@tophat.com), the in app support button, or by calling 1-888-663-5491.
- Peer evaluation feedback must be submitted via CATME by the time the online survey closes (typically 3-4 days after deliverables are submitted). CATME surveys will not be redistributed for a group under any circumstances.
- Dr. Cooper may ask a student group for a “long-form” peer evaluation to clarify CATME scores if (1) one or more team members received a peer evaluation score more than 15% below the group average and (2) little or no written explanation is given in CATME student comments for the low peer evaluation score(s). Student teams can avoid long-form evaluations by explaining peer evaluation scoring in comments through CATME.
- Peer evaluation scores may be adjusted at the discretion of the instructor in circumstances where it appears inappropriate team dynamics have prevented accurate peer evaluations.
- The instructor may assign a grade of zero for a student’s deliverable if it is apparent the student did not meaningfully contribute to the group assignment.
- It is prohibited for students in different groups to share graded, ungraded or in-progress course materials (including, but not limited to, reports, notes from Q&A sessions, homework, etc.) in any format. Giving another student(s) any course materials without instructor consent is considered "Aiding and Abetting" cheating as per NCSU's Code of Student Conduct. In other words, if a student in another group is found to have plagiarized your report, used your notes from Q&A sessions, etc., your grades will be eligible to be reduced by the Office of Student Conduct (OSC). If this occurs after you complete CHE 450, your grade is eligible to be retroactively reduced (possibly to an F, which would void your CBE degree if you've already graduated).

- Artificial intelligence (AI) language tools, such as (but not limited to) ChatGPT, Dall-e, etc. may not be used for course assignments except as explicitly authorized by the instructor. Unacceptable uses include (but are not limited to) the following actions:
 - Submitting all or any part of an assignment statement to an AI tool
 - Using AI to write a draft of a writing assignment
 - Using AI to write entire sentences, paragraphs, or papers to complete class assignments

The use of AI tools is permitted in this course for the following activities:

- Brainstorming and refining your ideas
- Fine-tuning research questions
- Finding information on your topic
- Drafting an outline to organize thoughts
- Checking grammar and style

You are responsible for the information you submit based on an AI query (for instance, that it does not violate intellectual property laws, or contain misinformation or unethical content). Your use of AI tools must be properly documented and cited in order to stay within university policies on academic honesty. For example, “Describe the symbolism of the green light in the book *The Great Gatsby* by F. Scott Fitzgerald” prompt. ChatGPT, Feb 13, 2024 version, OpenAI, chat.openai.com/chat.” Any assignment that is found to have used generative AI tools in unauthorized ways is a violation of academic misconduct. When in doubt about permitted usage, please ask the instructor for clarification.

- Coordinating with other students, or individuals outside the course, on any assignments and/or exams in any way can result in an academic integrity violation with the university. Unauthorized aid additionally includes uploading or accessing on-line solutions (e.g. Chegg), whether that involves posting problem solutions, accessing posted solutions, or paying someone to assist in solving problems. All cases of academic misconduct will be submitted to the Office of Student Conduct. Students found guilty of academic misconduct will be subject to, at a minimum, a zero on the assignment in question, up to a failing grade or a grade of incomplete (IN) for the course, depending on the nature of the violation. In addition, if you are found guilty of academic misconduct in the course, you will be on academic integrity probation for the remainder of your years at NCSU and may be required to report your violation on future professional school applications.
- All electronic and hard-copy documents (including but not limited to: copies of lecture notes, homework assignments and solutions, videos, exam assignments and solutions, graded/ungraded reports, handouts provided by the instructor, etc.) made available to students in this course are copyrighted by Dr. Matthew Cooper and intended only for students’ personal use. You are not allowed to share content of the class with any third parties, including any person not signed up for the course; a personal, public, or commercial website; or any news or advertising media. To be clear, you are not permitted to upload any content from this course to the web in any form, including but

not limited to Chegg, Course Hero, Coursera, Google Drive, etc. If you post course content, you risk violating the instructor's intellectual property rights as well as NCSU's Code of Student Conduct. All cases of academic misconduct will be submitted to the Office of Student Conduct. Students found guilty of academic misconduct will be subject to, at a minimum, a zero on the assignment in question, up to a failing grade or a grade of incomplete (IN) for the course, depending on the nature of the violation. In addition, if you are found guilty of academic misconduct in the course, you will be on academic integrity probation for the remainder of your years at NCSU and may be required to report your violation on future professional school applications.

- It is prohibited to make audio or video recordings of conversations with the course instructor outside of the scheduled lecture period.
- Students who decide to take this course as "pass/fail" at any point during the semester will not be permitted to participate in the capstone portion of the course.
- Office hours will not be held on dates when classes are not in session (e.g. on holidays, due to inclement weather).
- If you believe that an error was made in grading homework or exams, write a short justification of your claim for the particular assignment and email it to Dr. Cooper. The TA or instructor will review your concern and respond to you directly. The "statute of limitations" for submitting such claims is one week after the assignment is returned.
- In the event a student misses one exam for any reason, they have the option to take a comprehensive make-up exam in its place at the end of the semester, after the last week of classes. Make-up exams submitted by students who missed an exam for a university-approved reason are eligible for full credit. Make-up exams submitted by students who missed an exam for a non-university-approved reason (e.g. flat tire, overslept, etc.) will be penalized 30 points. If a student misses more than one exam, any additional exams missed for a non-university-approved reason will receive a grade of zero; this policy applies even if the first missed exam was for a university-approved reason.
- Cell phones, laptops, tablet computers, etc. must be turned off and put away during exams. If a student plans to take notes in digital format using a tablet computer, these notes must be printed in order to be used during exams.
- All students registered with DRO should make arrangements to take tests at the DRO testing center and the instructor should be notified in advance of any arrangements for testing. Students who are registered with DRO but do not make reservations with the DRO office more than three business days in advance of the test date can still inquire about the possibility of receiving their accommodations, but should plan to test with the rest of the class and without accommodations. DRO space (and faculty availability) is often minimal ahead of the three-day online scheduling cut-off so students should make all of their reservations as early in the semester as possible.
- Completion of the 16 SACHE modules specified on the CHE 450 Moodle site is required for graduation with a ChE degree at NCSU; i.e., the specified SACHE modules must be completed by the end of the CHE 451 course in the spring. SACHE modules are free for undergraduate AICHE student members. You must join

the national AIChE (membership is also free!) in order to access these modules (being a member of the local NCSU student chapter does not give you access). To join national AIChE, go to <https://www.aiche.org/community/students/student-membership>. Once completing the specified SACHE modules, save your certificate of completion for each module to upload to submission links in CHE 451.

- Bonus points may be awarded by the instructor for extraordinary achievement of a student or a team.
- There will be no final examination; the Fall Project Status Report deliverable represents the culmination of the fall semester of the program's capstone design sequence.

14. Inclusivity Statements

- Religious/Cultural Observance: Persons who have religious or cultural observances that coincide with this class should let the instructor know in writing via e-mail at least one week prior to the conflict to request any accommodations.
- Title IX Statement: Title IX makes it clear that violence and harassment based on sex or gender is a Civil Rights offense, subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race or national origin. If you or someone you know has been harassed or assaulted, you can find the appropriate resources here: <https://diversity.ncsu.edu/title-ix/>
- Inclusive Learning: At NCSU, administrators, faculty, and staff are committed to the creation and maintenance of "inclusive learning" spaces, where you shall be treated with respect and dignity and where all individuals are provided equitable opportunity to participate, contribute, and succeed. In this course, all students are welcome regardless of race/ethnicity, gender identities, gender expressions, sexual orientation, socio-economic status, age, disabilities, religion, regional background, Veteran status, citizenship status, nationality and other diverse identities that we each bring to class. The success of an inclusive classroom relies on the participation, support, and understanding of you and your peers. We encourage you to speak up and share your views, but also understand that you are doing so in a learning environment in which we all are expected to engage respectfully and with regard to the dignity of all others.
- Trans-Inclusive Statement: In an effort to affirm and respect the identities of transgender students in the classroom and beyond, please contact the instructor if you wish to be referred to using a name and/or pronouns other than what is listed in the student directory.

15. NCSU Guidance and Resources

Due to the COVID-19 pandemic, public health measures continue to be implemented across campus. Students should stay current with these practices and expectations through the [Protect the Pack](https://www.ncsu.edu/coronavirus/) website (<https://www.ncsu.edu/coronavirus/>). The sections below provide expectations and conduct related to COVID-19 issues.

Health and Participation in Class

We are most concerned about your health and the health of your classmates and instructors/TAs.

- If you test positive for COVID-19, or are told by a healthcare provider that you are presumed positive for the virus, you should not attend any hybrid or face-to-face (F2F) classes and work with your instructor on any adjustments necessary; also follow other university guidelines, including self-reporting ([Coronavirus Self Reporting](#)): Self-reporting is not only to help provide support to you, but also to assist in contact tracing for containing the spread of the virus.
- If you feel unwell, even if you have not been knowingly exposed to COVID-19, please do not come to a F2F class or activity.
- If you are in quarantine, have been notified that you may have been exposed to COVID-19, or have a personal or family situation related to COVID-19 that prevents you from attending this course in person (or synchronously), please connect with your instructor to make alternative plans, as necessary.
- If you need to make a request for an academic consideration related to COVID-19, such as a discussion about possible options for remote learning, please talk with your instructor.

Health and Well-Being Resources

These are difficult times, and academic and personal stress are natural results. Everyone is encouraged to [take care of themselves](#) and their peers. If you need additional support, there are many resources on campus to help you:

- Counseling Center ([NCSU Counseling Center](#))
- Student Health Services ([Health Services | Student](#))
- If the personal behavior of a classmate concerns or worries you, either for the classmate's well-being or yours, we encourage you to report this behavior to the NC State CARES team: ([Share a Concern](#)).
- If you or someone you know are experiencing food, housing or financial insecurity, please see the Pack Essentials Program ([Pack Essentials](#)).

Community Standards related to COVID-19

We are all responsible for protecting ourselves and our community. Please see the [community standards](#) (which have been updated for 2021) and Rule 04.21.01 regarding Personal Safety Requirements Related to COVID-19 [RUL 04.21.01 – Personal Safety Requirements Related to COVID-19 – Policies, Regulations & Rules](#)

Course Expectations Related to COVID-19:

- **Course Attendance:** NC State attendance policies can be found at: [REG 02.20.03 – Attendance Regulations – Policies, Regulations & Rules](#). Please refer to the course's attendance, absence, and deadline policies for additional details. If you are quarantined or otherwise need to miss class because you have been advised that you may have been exposed to COVID-19, you should not be penalized regarding attendance or class participation. However, you will be expected to develop a plan to keep up with your coursework during any such absences. If you become ill with COVID-19, you should follow the steps outlined in the health and participation section above. COVID 19-related absences will be considered excused; documentation need only involve communication with your instructor.

- **Technology Requirements:** This course may require particular technologies to complete coursework. Be sure to review the syllabus for these expectations, and see the [syllabus technical requirements](#) for your course. If you need access to additional technological support, please contact the Libraries' Technology Lending Service: ([Technology Lending](#)).

Course Delivery Changes Related to COVID-19

Please be aware that the situation regarding COVID-19 is frequently changing, and the delivery mode of this course could change accordingly, including from in-person to remote. Regardless of the delivery method, we will strive to provide a high-quality learning experience.

NO LONGER AVAILABLE - Grading/Scheduling Changing Options Related to COVID-19

Two policies, (1) enhanced S/U Grading Option, and (2) Late Course Drops, were put in place at the beginning of the COVID-19 pandemic to ease student stress and promote course completion. Those two policies *have been discontinued* and thus are no longer available to students.

For situations where relief may be needed, and depending on the details, students should explore the applicability of an “incomplete” grade in the course. If you are experiencing difficult or extenuating circumstances, you should discuss possible options with your instructor and your academic advisor.

Need Help?

If you find yourself in a place where you need help, academically or otherwise, please review these [Step-by-Step Help Topics](#).

Other Important Resources

- **Keep Learning:** [Keep Learning](#)
- **Protect the Pack FAQs:** [Frequently Asked Questions](#) | [Protect the Pack](#)
- **NC State Protect the Pack Resources for Students:** [Resources for Students](#) | [Protect the Pack](#)
- **Academic Success Center** (tutoring, drop in advising, career and wellness advising): [Academic Success Center](#).
- **NC State Keep Learning, tips for students opting to take courses remotely:** [Keep Learning Tips for Remote Learning](#)
- **Introduction to Zoom for students:** <https://youtu.be/5LbPzzPbYEW>
- **Learning with Moodle, a student's guide to using Moodle:** <https://moodle-projects.wolfware.ncsu.edu/course/view.php?id=226>
- **NC State Libraries** [Technology Lending Program](#)

16. Additional Required Text for NCSU Syllabi

- **Academic Integrity:** Students are required to comply with the university policy on academic integrity found in the Code of Student Conduct found at <http://policies.ncsu.edu/policy/pol-11-35-01>. It is the instructor's understanding and expectation that the

student's signature on any test or assignment means that the student contributed to the assignment in question (if a group assignment) and that they neither gave nor received unauthorized aid. Authorized aid on an individual assignment includes discussing the interpretation of the problem statement, sharing ideas or approaches for solving the problem, and explaining concepts involved in the problem. Any other aid would be unauthorized and a violation of the academic integrity policy. Unauthorized aid additionally includes accessing on-line solutions, whether that be complete or partial on-line copies of textbook problem solutions or paying someone to complete your homework. In addition, *any computer work submitted must be completed on your own personal computer or from your own @ncsu.edu account to avoid confusion about the origin of the file, and no sharing of files in any way is allowed.* All cases of academic misconduct will be submitted to the Office of Student Conduct. Students found guilty of academic misconduct will be subject to, at a minimum, a zero on the assignment in question, up to a zero for that course component (e.g. a zero for the homework portion of the final grade), or a failing grade in the course, depending on the nature of the violation. In addition, if you are found guilty of academic misconduct in the course, you will be on academic integrity probation for the remainder of your years at NCSU and may be required to report your violation on future professional school applications.

- **Students With Disabilities:** Reasonable accommodations will be made for students with verifiable disabilities. In order to take advantage of available accommodations, students must register with the Disability Services Office at Suite 2221, Student Health Center, Campus Box 7509, 919-515-7653. For more information on NC State's policy on working with students with disabilities, please see the **Academic Accommodations for Students with Disabilities Regulation (REG02.20.01)**(<http://policies.ncsu.edu/regulation/reg-02-20-01>)
- **Attendance Regulation:** <http://policies.ncsu.edu/regulation/reg-02-20-03>
- **Equal Opportunity:** NC State University provides equality of opportunity in education and employment for all students and employees. Accordingly, NC State affirms its commitment to maintain a work environment for all employees and an academic environment for all students that is free from all forms of discrimination. Discrimination based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation is a violation of state and federal law and/or NC State University policy and will not be tolerated. Harassment of any person (either in the form of quid pro quo or creation of a hostile environment) based on race, color, religion, creed, sex, national origin, age, disability, veteran status, or sexual orientation also is a violation of state and federal law and/or NC State University policy and will not be tolerated. Retaliation against any person who complains about discrimination is also prohibited. NC State's policies and regulations covering discrimination, harassment, and retaliation may be accessed at <http://policies.ncsu.edu/policy/pol-04-25-05> or http://www.ncsu.edu/equal_op/. Any person who feels that he or she has been the subject of prohibited discrimination, harassment, or retaliation should contact the Office for Equal Opportunity (OEO) at 919-515-3148.
- **NC State Policies, Regulations, and Rules (PRR):** Students are responsible for reviewing the NC State University PRR's located at <http://oucc.ncsu.edu/course-rights-and-responsibilities> which pertains to their course rights and responsibilities.

- **Classroom Capture:** Please be advised this course is being recorded for current and potential future educational purposes. By your continued participation in this recorded course, you are providing your permission to be recorded.
- **ClassEval:** Course and instructor evaluations: Online class evaluations will be available for students to complete during the last two weeks of class. Students will receive an email message directing them to a website where they can login using their Unity ID and complete evaluations. All evaluations are confidential; instructors will never know how any one student responded to any question, and students will never know the ratings for any particular instructors.
 - Evaluation website: <https://classeval.ncsu.edu>
 - More information: <http://www2.acs.ncsu.edu/UPA/classeval/index.htm>
 - Schedule: Online class evaluations will be available for students to complete during the last 2 weeks of class and become unavailable before finals begin.